ANTIBIOTIC MONOGRAPH

Piperacillin

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1. Classification

- Ureidopenicillin
- β-lactam antibiotic

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2. Mechanism of Action

- Inhibits **penicillin-binding proteins (PBPs)**
- Disrupts bacterial cell wall synthesis

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3. Spectrum of Activity

Strong activity:

- Gram-negatives:
  - *Pseudomonas aeruginosa*
  - *Enterobacterales*

Moderate:

- Gram-positives (less than penicillin G)

Weak:

- Anaerobes (improved with tazobactam)

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4. Limitations

- Highly susceptible to β-lactamases
  → Hence used clinically as:
- Piperacillin tazobactam

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5. Resistance

- β-lactamases
- Efflux pumps (Pseudomonas)
- Porin loss

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6. Clinical Role

- Severe hospital infections
- Sepsis
- Polymicrobial infections

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7. Key Insight

Piperacillin alone is pharmacologically valuable, but clinically obsolete without β-lactamase inhibition.

Cefiderocol

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1. Classification

- Siderophore cephalosporin

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2. Mechanism of Action

- Binds iron → actively transported into bacteria
- Then inhibits cell wall synthesis

Key innovation:

"Trojan horse" strategy to bypass permeability barriers

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3. Spectrum

- Highly active against:
  - *Acinetobacter baumannii*
  - *Pseudomonas aeruginosa*
  - Carbapenem-resistant Enterobacterales

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4. Clinical Role

- MDR/XDR Gram-negative infections
- Particularly valuable for:
  - Carbapenem-resistant *Acinetobacter*

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5. Resistance

- Still emerging
- Mechanisms:
  - Iron transport mutations
  - β-lactamases (partial impact)

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6. Key Insight

Cefiderocol represents a mechanistic shift—overcoming resistance not by inhibition, but by **forced intracellular entry**.

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8. Pharmacodynamic Considerations (A. baumannii)

- High MIC variability
- Combination therapy often required
- Monotherapy failure possible in severe infections

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9. Global Resistance Trends

- Rapid emergence of **XDR and PDR strains**
- Increasing **colistin resistance**
- Endemic in many ICUs globally

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10. Laboratory Considerations

- Difficult susceptibility interpretation
- Heteroresistance common
- Molecular detection:
  - OXA genes

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11. Clinical Decision Points

1. Carbapenem-resistant strain

→ Avoid:

- Meropenem

→ Consider:

- Colistin
- Tigecycline
- Cefiderocol

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2. Severe infection

→ Use:

- Combination therapy

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3. Suspected outbreak

→ Immediate infection control critical

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12. Stewardship and Public Health Role

- Indicator of:
  - ICU hygiene failure
  - Antibiotic pressure

Control priority organism globally

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13. Key Insight (Takeaway)

Acinetobacter baumannii represents the extreme end of antimicrobial resistance, where therapeutic options are not just limited—but progressively collapsing.

Where This Is Going

We now have three distinct resistance paradigms:

Organism	Resistance Model
E. coli	Plasmid-driven (ESBL, carbapenemase)
Klebsiella pneumoniae	Clonal + plasmid amplification
Pseudomonas aeruginosa	Intrinsic + adaptive
Acinetobacter baumannii	Accumulative + extreme MDR

Penicillin

---

1. Identity and Classification

- **Class:** β-lactam antibiotic
- **Subclass:** Natural penicillins
- **Core agents:**
  - Penicillin G (parenteral)
  - Penicillin V (oral)

Historical significance:

- First widely used antibiotic (revolutionized medicine)

---

2. Chemical Structure and Origin

- Derived from *Penicillium* fungi
- Core structure:
  - β-lactam ring
  - Thiazolidine ring

Key feature:
The β-lactam ring is essential for antibacterial activity but also highly vulnerable to degradation (β-lactamases).

---

3. Mechanism of Action

- Binds **penicillin-binding proteins (PBPs)**
- Inhibits **transpeptidation** (cell wall cross-linking)
- Leads to:
  - Cell wall instability
  - Osmotic lysis

Nature of activity:

- Bactericidal
- Time-dependent killing (T > MIC)

---

4. Spectrum of Activity

A. Strong Activity

Gram-positive organisms

- Streptococcus pyogenes
- Streptococcus agalactiae
- Streptococcus pneumoniae *(variable resistance)*

Anaerobes

- Clostridium perfringens

B. Moderate / Selective Activity

- Enterococcus faecalis *(requires synergy with aminoglycosides)*

C. Limited Gram-negative Activity

- Neisseria gonorrhoeae *(historically; now largely resistant)*

D. Poor or No Activity

- Staphylococcus aureus *(most strains produce penicillinase)*
- Escherichia coli
- Klebsiella pneumoniae *(intrinsic resistance)*
- Pseudomonas aeruginosa

---

5. Pharmacokinetics

Penicillin G

- IV or IM administration
- Poor oral stability

Penicillin V

- Oral formulation
- Better acid stability

Distribution:

- Wide tissue distribution
- Limited CNS penetration (unless meningitis)

Elimination:

- Renal

---

6. Resistance Mechanisms

A. β-lactamase production (dominant)

- Penicillinase (especially in *Staphylococcus aureus*)

B. Altered PBPs

- Seen in:
  - Streptococcus pneumoniae
  - Enterococcus faecium

C. Reduced permeability (Gram-negatives)

---

7. Clinical Uses

- Streptococcal infections
- Syphilis (*Treponema pallidum*, not in your list but essential context)
- Clostridial infections
- Rheumatic fever prophylaxis

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8. Adverse Effects

Common

- Hypersensitivity reactions (rash, urticaria)

Severe

- Anaphylaxis
- Stevens–Johnson syndrome (rare)

Others

- Jarisch–Herxheimer reaction (in spirochetal infections)

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9. Drug Interactions

- Probenecid → increases penicillin levels
- Minimal cytochrome P450 interaction

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10. Laboratory Considerations

- Susceptibility testing via MIC or disk diffusion
- β-lactamase testing for staphylococci

---

11. Stewardship Perspective

- Narrow-spectrum → **highly desirable when effective**
- Minimal ecological disruption compared to broad-spectrum agents

Key issue:

Widespread resistance has limited its use despite ideal stewardship profile.

---

12. Key Organism Mapping Insight

- Excellent for:
  - Streptococci
  - Some anaerobes
- Ineffective for:

    - Most Gram-negatives
    - β-lactamase producers

---

13. Takeaway

Penicillin represents the ideal antibiotic—narrow, effective, and low in collateral damage—but has been largely outpaced by bacterial evolution through widespread β-lactamase production.

Ampicillin

---

1. Identity and Classification

- **Class:** β-lactam antibiotic
- **Subclass:** Aminopenicillin
- **Related drugs:** Amoxicillin (oral analogue)

Position in evolution:

Ampicillin represents the first major expansion of penicillin activity into Gram-negative organisms.

---

2. Chemical Structure and Origin

- Semisynthetic derivative of penicillin
- Modified side chain improves:
    - Gram-negative penetration
    - Acid stability (partial oral use possible)

---

3. Mechanism of Action

- Binds **penicillin-binding proteins (PBPs)**
- Inhibits bacterial cell wall synthesis

Activity profile:

- Bactericidal
- Time-dependent killing (T > MIC)

---

4. Spectrum of Activity

A. Gram-positive Activity

- Streptococcus pyogenes
- Streptococcus agalactiae
- Streptococcus pneumoniae *(variable)*

Key feature:

- Retains most penicillin activity against streptococci

B. Enterococci (Critical Role)

- Enterococcus faecalis *(primary target)*
- Enterococcus faecium *(often resistant)*

Important:

Ampicillin is a cornerstone drug for *E. faecalis*

C. Gram-negative Activity (Expanded vs Penicillin)

- Escherichia coli *(high resistance today)*
- Haemophilus influenzae

- Proteus mirabilis

D. Anaerobes

- Some activity, but not reliable for broad anaerobic coverage

---

E. Ineffective Against

- Klebsiella pneumoniae *(intrinsic resistance)*
- Pseudomonas aeruginosa
- Staphylococcus aureus *(β-lactamase producers)*

---

5. Pharmacokinetics

- Oral and IV forms available
- Moderate bioavailability

Distribution:

- Good tissue penetration
- CNS penetration in meningitis

Elimination:

- Renal

---

6. Resistance Mechanisms

A. β-lactamases (Primary Limitation)

- TEM-type enzymes common

B. Altered PBPs

- Seen in:
  - Streptococcus pneumoniae
  - Enterococcus faecium

C. Reduced permeability (Gram-negatives)

---

7. Clinical Uses

- Enterococcal infections
- Neonatal infections (e.g., sepsis, meningitis)
- Respiratory infections (historically)
- Listeria infections *(not in your list but essential context)*

---

8. Adverse Effects

- Rash (notably in viral infections like EBV)
- Hypersensitivity reactions
- Gastrointestinal upset

---

9. Drug Interactions

- Similar to penicillin
- Probenecid increases levels

---

10. Laboratory Considerations

- Susceptibility testing required due to high resistance
- β-lactamase detection critical in Gram-negatives

---

11. Stewardship Perspective

- Narrower than many modern agents → desirable
- However:

High resistance rates limit empiric use

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12. Key Organism Mapping Insight

- Strongest niche:
  - Enterococci
  - Some streptococci
- Weakness:
  - Rapidly degraded by β-lactamases
  - Poor reliability in Gram-negatives today

---

13. Takeaway

Ampicillin expanded the antibacterial horizon of penicillins, but its clinical utility has been eroded by widespread β-lactamase-mediated resistance.

Amoxicillin–Clavulanate

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1. Identity and Classification

- **Class:** β-lactam + β-lactamase inhibitor
- **Components:**
  - Amoxicillin (aminopenicillin)
  - Clavulanic acid (β-lactamase inhibitor)

---

2. Mechanism of Action

Amoxicillin:

- Inhibits PBPs → cell wall disruption

Clavulanate:

- Irreversibly inhibits many β-lactamases

Net effect:

Restores activity of amoxicillin against β-lactamase-producing organisms

---

3. Spectrum of Activity

A. Gram-positive

- Streptococcus pneumoniae
- Staphylococcus aureus *(MSSA only)*

B. Gram-negative

- Haemophilus influenzae
- Escherichia coli *(if non-ESBL)*
- Moraxella catarrhalis

C. Anaerobes (Strong Feature)

- Bacteroides fragilis
- Prevotella spp

D. Ineffective Against

- Pseudomonas aeruginosa
- Acinetobacter baumannii
- ESBL-producing organisms

---

4. Pharmacokinetics

- Oral formulation widely used
- Good tissue penetration

---

5. Resistance Mechanisms

A. ESBLs

- Not inhibited by clavulanate

B. AmpC β-lactamases

- Resistant to inhibition

C. Carbapenemases

- No activity

---

6. Clinical Uses

- Respiratory infections
- Skin and soft tissue infections
- Bite wounds
- Intra-abdominal infections (mild/moderate)

---

7. Adverse Effects

- Gastrointestinal upset
- Diarrhea (common)
- Cholestatic hepatitis (rare but notable)

---

8. Stewardship Perspective

- Broad-spectrum oral agent
- Overuse → drives:
    - ESBL selection
    - Gut microbiome disruption

---

9. Key Organism Mapping Insight

- Strength:
    - β-lactamase-producing community pathogens
    - Anaerobes
- Weakness:
    - ESBL organisms

- Hospital-acquired Gram-negatives

---

10. Takeaway

Amoxicillin–clavulanate represents the first major pharmacological solution to β-lactamase-mediated resistance, but is ineffective against modern high-level resistance mechanisms such as ESBLs and carbapenemases.

Ampicillin–Sulbactam

---

1. Identity and Classification

- **Class:** β-lactam + β-lactamase inhibitor
- **Components:**
  - Ampicillin (aminopenicillin)
  - Sulbactam (β-lactamase inhibitor with intrinsic activity)

---

2. Mechanism of Action

Ampicillin:

- Inhibits PBPs → cell wall synthesis disruption

Sulbactam:

- Inhibits β-lactamases
- **Unique feature:**

Direct antibacterial activity against *Acinetobacter spp.*

---

3. Spectrum of Activity

A. Gram-positive

- Streptococcus pneumoniae
- Enterococcus faecalis

B. Gram-negative

- Escherichia coli *(non-ESBL)*
- Haemophilus influenzae

C. Critical Feature

- Acinetobacter baumannii *(key indication)*

D. Anaerobes

- Bacteroides fragilis

E. Ineffective Against

- Pseudomonas aeruginosa
- ESBL-producing organisms

---

4. Pharmacokinetics

- IV administration
- Good tissue penetration

---

5. Resistance Mechanisms

- ESBLs (not inhibited effectively)

- Carbapenemases
- Efflux and permeability changes (Acinetobacter)

---

6. Clinical Uses

- Intra-abdominal infections
- Skin and soft tissue infections
- **Acinetobacter infections (important niche)**

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7. Key Insight

Sulbactam transforms this combination into a targeted therapy for *Acinetobacter*, not just a β-lactamase workaround.

Piperacillin tazobactam

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1. Identity and Classification

- Class: β-lactam + β-lactamase inhibitor
- Components:
  - Piperacillin (ureidopenicillin)
  - Tazobactam

Position:

One of the most widely used broad-spectrum empiric antibiotics in hospital medicine.

---

2. Chemical and Structural Features

- Piperacillin: extended-spectrum penicillin with enhanced Gram-negative penetration
- Tazobactam: structurally similar to clavulanate but with broader inhibition

---

3. Mechanism of Action

Piperacillin:

- Binds PBPs → inhibits cell wall synthesis

Tazobactam:

- Irreversibly inhibits many class A β-lactamases

Net effect:

- Extends activity against β-lactamase-producing organisms

---

4. Spectrum of Activity

A. Gram-positive

- Streptococcus pneumoniae
- Enterococcus faecalis

Note:

- No reliable MRSA coverage

B. Gram-negative (Major Strength)

- Pseudomonas aeruginosa
- Escherichia coli
- Klebsiella pneumoniae
- Enterobacter cloacae

C. Anaerobes

- Bacteroides fragilis
- Prevotella spp

D. Ineffective Against

- ESBL-producing Enterobacterales (unreliable)
- Carbapenem-resistant organisms
- Acinetobacter baumannii *(variable/poor)*

---

5. Pharmacokinetics and Pharmacodynamics

- IV administration
- Wide tissue distribution

PK/PD:

- Time-dependent killing
- Requires prolonged infusion in severe infections

Elimination:

- Renal

---

6. Resistance Mechanisms

A. ESBLs

- Not effectively inhibited

B. AmpC β-lactamases

- Inducible resistance (Enterobacter spp.)

C. Efflux and porin mutations

- Especially in *Pseudomonas*

---

7. Clinical Uses

- Sepsis (empiric therapy)
- Intra-abdominal infections
- Hospital-acquired pneumonia
- Febrile neutropenia

---

8. Adverse Effects

- Hypersensitivity reactions
- Nephrotoxicity (especially with vancomycin)
- Electrolyte disturbances (hypokalemia)

---

9. Drug Interactions

- Increased nephrotoxicity with:
    - Vancomycin

---

10. Laboratory Considerations

- Susceptibility testing critical due to rising resistance
- ESBL detection influences interpretation

11. Stewardship Perspective

- High ecological impact
- Overuse strongly linked to:
  - ESBL emergence
  - Selection of resistant *Pseudomonas*

12. Key Organism Mapping Insight

- Strength:
  - Broad empiric coverage including anaerobes
- Weakness:
  - Fails against advanced resistance (ESBL, CRE)

13. Takeaway

Piperacillin–tazobactam is a cornerstone empiric agent, but its clinical reliability is increasingly undermined by evolving β-lactamase-mediated resistance.

Cephalosporins

Core Concept

- β-lactams with **generation-based spectrum evolution**

Generation	Trend
Early	Gram-positive
Later	Gram-negative expansion
Advanced	Anti-MDR / anti-MRSA

Ceftriaxone

Ceftriaxone

Key Features

- 3rd-generation cephalosporin
- Long half-life (once daily dosing)

Spectrum

- Escherichia coli
- Klebsiella pneumoniae
- Streptococcus pneumoniae

Limitations

- No activity vs:
  - Pseudomonas aeruginosa
- Destroyed by ESBL

Insight

Backbone of modern medicine—but heavily compromised by ESBL spread.

Ceftazidime

Key Feature

- Anti-pseudomonal cephalosporin

Spectrum

- Pseudomonas aeruginosa

Weakness

- Poor Gram-positive activity

Cefepime

Key Feature

- 4th-generation cephalosporin

Spectrum

- Broad Gram-negative + Gram-positive
- Covers:
    - Pseudomonas aeruginosa

Resistance

- ESBL → failure

Cefoxitin

Class

- Cephamycin

Key Feature

- Better anaerobic activity

Use

- Intra-abdominal infections

Ceftaroline

Unique Feature

Only cephalosporin active against MRSA

Spectrum

- Staphylococcus aureus (MRSA)

Ceftazidime–Avibactam

Ceftazidime avibactam

Key Role

- Active against:
  - KPC
  - OXA-48

Limitation

- Not active against MBL alone

Ceftolozane–Tazobactam

Key Feature

One of the most potent drugs for MDR *Pseudomonas*

Carbapenems

Carbapenems

Core Properties

- Broadest β-lactams
- Resistant to most β-lactamases

Meropenem / Imipenem / Doripenem / Ertapenem

Spectrum

- Gram-positive
- Gram-negative
- Anaerobes

Key Differences

Drug	Key Feature
Meropenem	CNS penetration
Imipenem	seizure risk
Doripenem	anti-Pseudomonas
Ertapenem	no Pseudomonas

Resistance

- Carbapenemases (KPC, NDM)

Meropenem–Vaborbactam

Key Role

- KPC-producing organisms

Aztreonam

Unique Feature

- Monobactam
- Safe in penicillin allergy

Spectrum

- Gram-negative only

Aztreonam–Avibactam

Critical Role

Only reliable option for **NDM-producing organisms**

Aminoglycosides

Amikacin / Gentamicin

Key Features

- Concentration-dependent killing
- Strong Gram-negative activity

Limitations

- Nephrotoxicity
- Ototoxicity

Fluoroquinolones

Ciprofloxacin / Levofloxacin / Moxifloxacin

Mechanism

- DNA gyrase inhibition

Key Issue

Rapid resistance development

Vancomycin / Linezolid / Daptomycin

Spectrum

- Gram-positive only

Roles

- MRSA
- VRE

Colistin

Role

- Last-resort for MDR Gram-negatives

Toxicity

- Nephrotoxicity

Tigecycline

Spectrum

- Very broad (except Pseudomonas)

Metronidazole

Spectrum

- Anaerobes
- Clostridioides difficile

Trimethoprim–Sulfamethoxazole

Mechanism

- Folate pathway inhibition

Ceftriaxone

---

1. Identity and Classification

- **Class:** β-lactam
- **Subclass:** 3rd-generation cephalosporin

Defining feature:

Long half-life → once-daily dosing

---

2. Chemical Structure

- Cephalosporin nucleus with enhanced Gram-negative penetration

---

3. Mechanism of Action

- Inhibits PBPs → disrupts cell wall synthesis

4. Spectrum of Activity

A. Gram-negative

- Escherichia coli
- Klebsiella pneumoniae
- Haemophilus influenzae

B. Gram-positive

- Streptococcus pneumoniae

C. Notable Activity

- Neisseria gonorrhoeae

D. Ineffective Against

- Pseudomonas aeruginosa
- Enterococcus spp

---

5. Pharmacokinetics

- Long half-life (~8 hours)
- Dual elimination:
  - Renal
  - Biliary

---

6. Resistance Mechanisms

A. ESBLs (Major)

- CTX-M enzymes

B. AmpC β-lactamases

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7. Clinical Uses

- Community-acquired pneumonia
- Sepsis
- Meningitis
- Gonorrhea

---

8. Adverse Effects

- Biliary sludging
- Hypersensitivity
- Rare hemolysis

---

9. Stewardship Perspective

- Widely overused → major driver of ESBL resistance

---

10. Key Insight

Ceftriaxone is one of the most clinically valuable antibiotics—but also one of the most compromised by global resistance.

Cefepime

---

1. Identity and Classification

- **Class:** β-lactam
- **Generation:** 4th-generation cephalosporin

---

2. Mechanism

- PBP inhibition

---

3. Spectrum

Strong Gram-negative:

- Pseudomonas aeruginosa
- Enterobacter cloacae

Gram-positive:

- Streptococcus spp

---

4. Key Feature

More stable against AmpC β-lactamases than earlier cephalosporins

---

5. Limitations

- ESBL → resistance
- No anaerobic coverage

---

6. Clinical Uses

- Severe hospital infections
- Neutropenic fever

---

7. Adverse Effects

- Neurotoxicity (encephalopathy, seizures) in renal failure

---

8. Insight

Cefepime bridges traditional cephalosporins and carbapenems—but does not overcome ESBL resistance.

Meropenem

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1. Classification

- Carbapenem

---

2. Mechanism

- Broad PBP binding → potent bactericidal activity

---

3. Spectrum

Very broad:

- Escherichia coli
- Klebsiella pneumoniae
- Pseudomonas aeruginosa

- Bacteroides fragilis

---

4. Key Feature

Resistant to most β-lactamases (except carbapenemases)

---

5. Resistance

- KPC, NDM, OXA enzymes

---

6. Clinical Uses

- ESBL infections
- Severe sepsis
- Intra-abdominal infections

---

7. Adverse Effects

- Generally well tolerated
- Rare seizures

---

8. Stewardship Insight

Overuse leads directly to carbapenem-resistant organisms—the most dangerous AMR category.

Vancomycin

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1. Classification

- Glycopeptide

---

2. Mechanism

- Binds D-Ala-D-Ala → inhibits cell wall synthesis

---

3. Spectrum

Gram-positive only:

- Staphylococcus aureus (MRSA)
- Enterococcus faecalis

---

4. Resistance

- VRE (vanA, vanB genes)

---

5. Clinical Uses

- MRSA infections
- Severe Gram-positive infections

---

6. Adverse Effects

- Nephrotoxicity
- “Red man syndrome”

---

7. Insight

Vancomycin defines the transition into targeted Gram-positive resistance management.

Imipenem

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1. Identity and Classification

- **Class:** Carbapenem
- **Formulation:** Always co-administered with cilastatin

Key feature:

First clinically used carbapenem; extremely broad-spectrum

---

2. Chemical and Structural Features

- β-lactam with a carbapenem ring
- Highly resistant to most β-lactamases

Cilastatin role:

- Inhibits renal dehydropeptidase → prevents imipenem degradation

---

3. Mechanism of Action

- Binds multiple PBPs
- Inhibits cell wall synthesis

Effect:

- Rapid bactericidal activity

---

4. Spectrum of Activity

A. Gram-negative

- Escherichia coli
- Klebsiella pneumoniae
- Pseudomonas aeruginosa

B. Gram-positive

- Staphylococcus aureus (MSSA)
- Enterococcus faecalis

C. Anaerobes

- Bacteroides fragilis

D. Limitations

- No MRSA coverage
- Ineffective against carbapenem-resistant organisms

---

5. Pharmacokinetics

- IV administration
- Wide distribution

Elimination:

- Renal

---

6. Resistance Mechanisms

- Carbapenemases (KPC, NDM, OXA)
- Porin loss (especially in *Pseudomonas*)

7. Clinical Uses

- Severe polymicrobial infections
- Sepsis
- Intra-abdominal infections

---

8. Adverse Effects

- **Seizure risk (important distinguishing feature)**
- GI upset
- Hypersensitivity

---

9. Stewardship Insight

Imipenem's potency comes at a cost—its overuse directly accelerates carbapenem resistance.

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10. Takeaway

Imipenem established the carbapenem class as the ultimate β-lactam defense—now increasingly threatened by carbapenemases.

Ertapenem

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1. Identity and Classification

- **Class:** Carbapenem

Defining feature:

Narrower spectrum compared to other carbapenems

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2. Mechanism of Action

- PBP inhibition → bactericidal

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3. Spectrum of Activity

A. Strong activity

- Escherichia coli (including ESBL)
- Klebsiella pneumoniae
- Bacteroides fragilis

B. Key Limitation

- No activity against:
  - Pseudomonas aeruginosa
  - Acinetobacter baumannii

---

4. Pharmacokinetics

- Once-daily dosing
- High protein binding

---

5. Clinical Uses

- ESBL infections (non-Pseudomonas)
- Intra-abdominal infections
- Community-acquired infections

6. Stewardship Insight

Ertapenem is a "carbapenem-sparing carbapenem"—useful when Pseudomonas coverage is unnecessary.

7. Takeaway

Ertapenem trades spectrum for convenience, making it ideal for targeted ESBL therapy outside ICU settings.

Doripenem

1. Identity

- Carbapenem

2. Key Feature

Strong anti-*Pseudomonas* activity

3. Spectrum

- Pseudomonas aeruginosa
- Escherichia coli
- Klebsiella pneumoniae

4. Clinical Role

- Severe hospital infections

5. Limitation

- Reduced use due to clinical trial concerns in pneumonia

6. Takeaway

Pharmacologically strong but clinically less utilized compared to meropenem.

Aztreonam

1. Classification

- Monobactam

2. Mechanism

- PBP binding → cell wall inhibition

3. Spectrum

Gram-negative only:

- Pseudomonas aeruginosa
- Escherichia coli

No activity:

- Gram-positive
- Anaerobes

4. Unique Feature
Safe in penicillin allergy

---

5. Resistance

- ESBL → resistance
- MBL → preserved activity (important with avibactam)

---

6. Takeaway
Aztreonam is narrow but strategically critical in complex resistance scenarios.

Ciprofloxacin

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1. Classification

- Fluoroquinolone

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2. Mechanism

- Inhibits DNA gyrase and topoisomerase IV

---

3. Spectrum
Strong Gram-negative:

- Pseudomonas aeruginosa
- Escherichia coli

Limited Gram-positive

---

4. Pharmacodynamics

- Concentration-dependent
- AUC/MIC driven

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5. Resistance

- Target mutations
- Efflux pumps

---

6. Adverse Effects

- Tendon rupture
- QT prolongation
- CNS effects

---

7. Takeaway
Highly effective historically—but now heavily compromised by resistance.

Linezolid

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1. Classification

- Oxazolidinone

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2. Mechanism

- Inhibits 50S ribosomal subunit
- Blocks initiation of protein synthesis

---

3. Spectrum

- Staphylococcus aureus (MRSA)
- Enterococcus faecium (VRE)

---

4. Clinical Uses

- MRSA pneumonia
- VRE infections

---

5. Adverse Effects

- Bone marrow suppression
- Peripheral neuropathy
- Serotonin syndrome

---

6. Takeaway

Linezolid is a critical oral option for resistant Gram-positive infections.

Colistin

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1. Classification

- Polymyxin

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2. Mechanism

- Disrupts bacterial cell membrane

---

3. Spectrum

- Acinetobacter baumannii
- Pseudomonas aeruginosa
- CRE organisms

---

4. Role

- Last-resort therapy

---

5. Toxicity

- Nephrotoxicity
- Neurotoxicity

---

6. Resistance

- mcr genes

---

7. Takeaway

Colistin represents the final line of defense—yet even this is eroding.

Amikacin

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1. Identity and Classification

- **Class:** Aminoglycoside
- **Generation:** Semisynthetic derivative of kanamycin

Position in therapy:

Amikacin is the most **resistance-resilient aminoglycoside**, often retained when gentamicin fails.

---

2. Chemical and Structural Features

- Modified aminoglycoside structure designed to:
  - Resist enzymatic degradation
  - Preserve activity against resistant Gram-negatives

Key implication:

Structural modification directly translates into **clinical durability against resistance mechanisms**.

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3. Mechanism of Action

- Binds irreversibly to the **30S ribosomal subunit**
- Causes:
  - Misreading of mRNA
  - Premature termination of protein synthesis

Result:

- Production of defective proteins
- Rapid bacterial death

---

4. Pharmacodynamics

- **Concentration-dependent killing**
- Key parameter: **Cmax/MIC ratio**

Important features:

- Strong **post-antibiotic effect**
- Once-daily dosing often preferred

---

5. Spectrum of Activity

A. Gram-negative (Primary Strength)

- Pseudomonas aeruginosa
- Klebsiella pneumoniae
- Escherichia coli
- Acinetobacter baumannii

B. Gram-positive (Limited)

- Staphylococcus aureus *(only in combination therapy)*

C. No activity against:

- Anaerobes (oxygen-dependent uptake mechanism)

---

6. Resistance Mechanisms

A. Aminoglycoside-modifying enzymes (AMEs)

- Acetylation

- Phosphorylation
- Adenylation

Key advantage:

Amikacin resists many AMEs that inactivate gentamicin

B. 16S rRNA methylation

- High-level resistance (serious concern)

C. Reduced uptake

- Especially in biofilms or anaerobic conditions

---

7. Clinical Uses

- Severe Gram-negative infections
- Sepsis
- Combination therapy (e.g., with β-lactams)

Special role:

- Salvage therapy in MDR infections

---

8. Adverse Effects

Major toxicities:

- **Nephrotoxicity** (dose-related)
- **Ototoxicity** (irreversible hearing loss)

Monitoring required:

- Drug levels
- Renal function

---

9. Stewardship Perspective

- Highly effective but:
  - Toxicity limits use
  - Should be used in **targeted or combination therapy**

---

10. Takeaway

Amikacin represents the most durable aminoglycoside, retaining activity where others fail, but its clinical utility is constrained by toxicity and emerging high-level resistance mechanisms.

Gentamicin

---

1. Identity and Classification

- Aminoglycoside

Clinical position:

Historically one of the most widely used agents for Gram-negative infections.

---

2. Mechanism of Action

- Same as amikacin:
  - 30S ribosomal binding
  - Bactericidal

3. Spectrum of Activity

A. Gram-negative

- Escherichia coli
- Klebsiella pneumoniae

B. Gram-positive synergy

- Enterococcus faecalis
- Staphylococcus aureus

Important:

Used in combination to achieve bactericidal synergy

4. Resistance

- More susceptible than amikacin to enzymatic modification

5. Clinical Uses

- Sepsis
- Endocarditis (in combination)

6. Toxicity

- Same as amikacin
- Narrow therapeutic window

7. Key Insight

Gentamicin is highly effective but increasingly limited by resistance and toxicity.

Levofloxacin

1. Classification

- Fluoroquinolone

2. Mechanism

- DNA gyrase + topoisomerase IV inhibition

3. Spectrum

A. Gram-negative

- Escherichia coli
- Pseudomonas aeruginosa

B. Gram-positive (better than ciprofloxacin)

- Streptococcus pneumoniae

4. Clinical Uses

- Pneumonia
- UTIs
- Skin infections

5. Resistance

- Rapid selection under pressure

6. Adverse Effects

- QT prolongation
- Tendon rupture
- CNS effects

7. Insight

Levofloxacin bridges Gram-negative and respiratory Gram-positive coverage—but resistance is rising quickly.

Moxifloxacin

1. Key Feature

Strongest Gram-positive and anaerobic activity among fluoroquinolones

2. Spectrum

- Streptococcus pneumoniae
- Anaerobes

3. Limitation

- Poor activity vs:
    - Pseudomonas aeruginosa

4. Insight

More “respiratory-focused” than ciprofloxacin or levofloxacin.

Tigecycline

1. Classification

- Glycylcycline (tetracycline derivative)

2. Mechanism

- 30S ribosomal inhibition

3. Spectrum

Very broad:

- Acinetobacter baumannii
- Staphylococcus aureus (MRSA)
- Enterococcus faecium (VRE)

Limitation:

- No activity against:
    - Pseudomonas aeruginosa

4. Clinical Uses

- Complicated intra-abdominal infections
- MDR infections

---

5. Key Concern

- Low serum levels → not ideal for bacteremia

---

6. Insight

Tigecycline is broad but pharmacokinetically limited.

Metronidazole

---

1. Classification

- Nitroimidazole

---

2. Mechanism

- Reduced in anaerobic conditions → DNA damage

---

3. Spectrum

Anaerobes:

- Bacteroides fragilis
- Clostridioides difficile

---

4. Key Feature

Only works in **anaerobic environments**

---

5. Clinical Uses

- Intra-abdominal infections
- C. difficile infection

---

6. Adverse Effects

- Metallic taste
- Disulfiram-like reaction

---

7. Insight

Metronidazole is the backbone of anaerobic infection treatment.

Clindamycin

---

1. Identity and Classification

- **Class:** Lincosamide
- Structurally related to lincomycin

Clinical niche:

Particularly valuable for **anaerobic infections and toxin suppression**

---

2. Mechanism of Action

- Binds to the **50S ribosomal subunit**

- Inhibits protein synthesis

Effect:

- Primarily bacteriostatic
- Can be bactericidal at high concentrations for some organisms

---

3. Spectrum of Activity

A. Gram-positive

- Staphylococcus aureus *(including some MRSA strains)*
- Streptococcus pyogenes

B. Anaerobes (Major Strength)

- Bacteroides fragilis *(variable resistance increasing)*
- Clostridium perfringens

C. Limited / No Activity

- Escherichia coli
- Pseudomonas aeruginosa

---

4. Unique Functional Role

Toxin suppression

Clindamycin reduces toxin production in:

- Streptococcus pyogenes
- Staphylococcus aureus

This makes it critical in:

- Necrotizing infections
- Toxic shock syndromes

---

5. Resistance Mechanisms

A. MLS_B resistance

- Methylation of ribosomal binding site

Important phenomenon:

- **Inducible resistance (D-test positive)**

---

6. Clinical Uses

- Skin and soft tissue infections
- Anaerobic infections
- Necrotizing fasciitis (adjunct)

---

7. Adverse Effects

Major:

- **Clostridioides difficile infection (highest risk among antibiotics)**

---

8. Takeaway

Clindamycin is a uniquely strategic antibiotic—not just antimicrobial, but anti-toxin—yet its use is constrained by high risk of C. difficile infection.

Azithromycin

---

1. Classification

- Macrolide

---

2. Mechanism

- Binds 50S ribosomal subunit
- Inhibits protein synthesis

---

3. Spectrum

A. Gram-positive

- Streptococcus pneumoniae

B. Gram-negative (limited but important)

- Haemophilus influenzae

C. Atypicals (major strength)

- Not in your list but critical:
  - Mycoplasma
  - Chlamydia
  - Legionella

---

4. Pharmacokinetics

- Extremely long half-life
- High intracellular penetration

---

5. Clinical Uses

- Respiratory infections
- Sexually transmitted infections
- Atypical pneumonia

---

6. Resistance

- Ribosomal methylation
- Efflux pumps

---

7. Insight

Azithromycin is less about potency and more about **distribution and intracellular activity**.

Erythromycin

---

1. Classification

- First-generation macrolide

---

2. Spectrum

- Similar to azithromycin but:
  - Less Gram-negative coverage

- More side effects

---

3. Key Limitation

- GI intolerance (motilin receptor stimulation)

---

4. Clinical Role

- Largely replaced by newer macrolides

---

5. Insight

Erythromycin is historically important but clinically superseded.

Clarithromycin

---

1. Key Feature

- Improved stability vs erythromycin

---

2. Spectrum

- Similar to azithromycin
- Better activity vs some Gram-positives

---

3. Clinical Uses

- Respiratory infections
- H. pylori therapy

---

4. Key Issue

- Drug interactions (CYP450 inhibition)

---

5. Insight

Clarithromycin sits between erythromycin and azithromycin in both potency and tolerability.

Tetracycline

---

1. Classification

- Tetracycline class

---

2. Mechanism

- 30S ribosomal inhibition

---

3. Spectrum

Broad:

- Gram-positive
- Gram-negative
- Atypicals

---

4. Limitations

- Widespread resistance

- Reduced use today

---

5. Adverse Effects

- Tooth discoloration
- Photosensitivity

---

6. Insight

Once broad and powerful, now largely limited by resistance.

Minocycline

---

1. Classification

- Second-generation tetracycline

---

2. Key Feature

Improved activity against MDR organisms

---

3. Spectrum

- Acinetobacter baumannii
- Staphylococcus aureus

---

4. Clinical Role

- MDR infections
- Skin infections

---

5. Insight

Minocycline is a revival drug in the MDR era.

Teicoplanin

---

1. Classification

- Glycopeptide

---

2. Mechanism

- Same as vancomycin
- Cell wall inhibition

---

3. Spectrum

- Staphylococcus aureus
- Enterococcus faecalis

---

4. Key Advantage

Less nephrotoxic than vancomycin

---

5. Clinical Uses

- MRSA infections

- Gram-positive infections

---

6. Insight

Teicoplanin offers a safer alternative to vancomycin where available.

Quinupristin–Dalfopristin

---

1. Identity and Classification

- **Class:** Streptogramin combination
- Components:
  - Quinupristin (streptogramin B)
  - Dalfopristin (streptogramin A)

Key concept:

Synergistic dual binding → dramatically enhanced antibacterial effect

---

2. Mechanism of Action

- Both bind the **50S ribosomal subunit**
- Dalfopristin enhances quinupristin binding

Effect:

- Inhibits protein synthesis
- Converts bacteriostatic activity → **bactericidal effect**

---

3. Spectrum of Activity

A. Gram-positive (primary target)

- Staphylococcus aureus (including MRSA)
- Enterococcus faecium *(VRE — key indication)*

B. Limitation

- Not active against:
  - Enterococcus faecalis

---

4. Clinical Uses

- VRE (*E. faecium*) infections
- Resistant Gram-positive infections

---

5. Adverse Effects

- Myalgia and arthralgia
- Infusion-related pain
- Significant drug interactions (CYP3A4 inhibition)

---

6. Encyclopedic Takeaway

Quinupristin–dalfopristin is a highly specialized agent targeting resistant Gram-positive organisms, particularly VRE, but its clinical use is limited by toxicity and pharmacological complexity.

Trimethoprim–Sulfamethoxazole

---

1. Identity and Classification

- **Class:** Folate pathway inhibitors (combination therapy)

---

2. Mechanism of Action

Dual blockade:

- Sulfamethoxazole → inhibits dihydropteroate synthase
- Trimethoprim → inhibits dihydrofolate reductase

Result:

Sequential inhibition → synergistic bactericidal effect

---

3. Spectrum of Activity

A. Gram-negative

- Escherichia coli
- Klebsiella pneumoniae

B. Gram-positive

- Staphylococcus aureus *(including some MRSA)*

C. Special organisms

- Stenotrophomonas maltophilia *(key drug)*

---

4. Resistance Mechanisms

- Target enzyme mutations
- Plasmid-mediated resistance genes

---

5. Clinical Uses

- Urinary tract infections
- Skin infections
- Opportunistic infections

---

6. Adverse Effects

- Stevens–Johnson syndrome
- Bone marrow suppression
- Hyperkalemia

---

7. Insight

TMP-SMX remains clinically valuable due to oral availability and broad coverage, but resistance is widespread.

Cefixime

---

1. Classification

- Oral 3rd-generation cephalosporin

---

2. Spectrum

- Escherichia coli
- Klebsiella pneumoniae
- Haemophilus influenzae

3. Limitation

- No activity vs:
  - Pseudomonas aeruginosa
- ESBL → resistance

4. Clinical Uses

- UTIs
- Respiratory infections

5. Insight

Cefixime is convenient but increasingly unreliable due to resistance.

Cefpodoxime

1. Classification

- Oral cephalosporin

2. Spectrum

- Similar to cefixime

3. Clinical Role

- Step-down therapy

4. Limitation

- ESBL resistance

5. Insight

Cefpodoxime functions mainly as a continuation therapy rather than frontline treatment.

Ceftibuten

1. Classification

- Oral 3rd-generation cephalosporin

2. Spectrum

- Gram-negative focused
- Limited Gram-positive

3. Limitation

- ESBL → ineffective

4. Insight

Ceftibuten is niche and overshadowed by resistance patterns.

Tebipenem

1. Identity and Classification

- **Class:** Carbapenem
- **Unique feature:** Oral formulation

---

2. Mechanism

- PBP inhibition

---

3. Spectrum

- Escherichia coli (including ESBL)
- Klebsiella pneumoniae

---

4. Limitation

- No reliable activity vs:
  - Pseudomonas aeruginosa

---

5. Clinical Importance

Represents a major shift toward **oral treatment of MDR infections**

---

6. Insight

Tebipenem could redefine outpatient management of resistant infections.

Cefoperazone–Sulbactam

---

1. Identity and Classification

- **Class:** 3rd-generation cephalosporin + β-lactamase inhibitor
- Components:
  - Cefoperazone
  - Sulbactam

Distinctive feature:

Combines cephalosporin activity with **intrinsic anti-*Acinetobacter*** effect from sulbactam

---

2. Mechanism of Action

Cefoperazone:

- PBP inhibition → cell wall synthesis disruption

Sulbactam:

- β-lactamase inhibition
- Direct antibacterial action (not just inhibitor)

---

3. Spectrum of Activity

A. Gram-negative

- Escherichia coli
- Klebsiella pneumoniae

B. Key Strength

- Acinetobacter baumannii

C. Gram-positive

- Streptococcus spp

D. Limitations

- ESBL-producing organisms
- Carbapenem-resistant strains (variable)

---

4. Pharmacokinetics

- Dual elimination:
  - Biliary (cefoperazone)
  - Renal (sulbactam)

---

5. Clinical Uses

- Hospital-acquired infections
- Intra-abdominal infections
- Acinetobacter infections

---

6. Adverse Effects

- Hypoprothrombinemia (vitamin K interference)
- Bleeding risk

---

7. Encyclopedic Takeaway

Cefoperazone–sulbactam is a strategically important agent in regions with high Acinetobacter prevalence, leveraging sulbactam's unique intrinsic activity.

Ceftaroline–Avibactam

---

1. Identity and Classification

- Advanced β-lactam/β-lactamase inhibitor combination
- Still largely investigational / limited availability

---

2. Mechanism

Ceftaroline:

- Binds PBP2a → active against MRSA

Avibactam:

- Inhibits:
  - KPC
  - OXA-48
  - Some class C β-lactamases

---

3. Spectrum

A. Gram-positive

- Staphylococcus aureus (MRSA)

B. Gram-negative

- ESBL producers

- Some carbapenemase producers

---

4. Significance

Attempts to unify:

- MRSA coverage
- MDR Gram-negative coverage

---

5. Limitation

- Not active against metallo-β-lactamases (NDM)

---

6. Insight

Represents the next evolutionary step in β-lactam design—combining Gram-positive and Gram-negative resistance coverage.

Ceftibuten–Avibactam

---

1. Identity

- Oral cephalosporin + avibactam

---

2. Key Feature

Designed for **oral treatment of ESBL infections**

---

3. Spectrum

- Escherichia coli (ESBL)
- Klebsiella pneumoniae

---

4. Clinical Importance

- Potential outpatient alternative to carbapenems

---

5. Insight

Could significantly reduce hospital admissions for MDR infections.

Gatifloxacin

---

1. Classification

- Fluoroquinolone

---

2. Spectrum

- Broad Gram-positive and Gram-negative

---

3. Major Limitation

- Dysglycemia (serious adverse effect)

---

4. Clinical Status

Largely withdrawn or restricted

---

5. Insight

Demonstrates how toxicity—not resistance—can limit antibiotic utility.

Oxacillin

---

1. Identity and Classification

- **Class:** Anti-staphylococcal penicillin

---

2. Mechanism

- PBP inhibition

---

3. Key Feature

Resistant to staphylococcal β-lactamases

---

4. Spectrum

Strong activity:

- Staphylococcus aureus (MSSA)

No activity:

- MRSA
- Gram-negatives

---

5. Clinical Uses

- MSSA infections (gold standard therapy)

---

6. Insight

Oxacillin defines the distinction between MSSA and MRSA in clinical microbiology.

Coverage Matrix (Antibiotic → Organism Activity Map)

A. High-Priority Gram-Negatives

Antibiotic	Escherichia coli	Klebsiella pneumoniae	Pseudomonas aeruginosa	Acinetobacter baumannii
Ceftriaxone	Variable / resistance common	Variable / resistance common	No activity	No activity
Cefepime	Variable / resistance common	Variable / resistance common	Reliable activity	No activity
Piperacillin–tazobactam	Variable / resistance common	Variable / resistance common	Reliable activity	No activity
Meropenem	Reliable activity	Reliable activity	Reliable activity	Variable / resistance common
Ertapenem	Reliable activity	Reliable activity	No activity	No activity
Ceftazidime–avibactam	Reliable activity	Reliable activity	Variable / resistance common	No activity
Ceftolozane–tazobactam	Variable / resistance common	Variable / resistance common	Reliable activity	No activity

Aztreonam	Variable / resistance common	Variable / resistance common	Reliable activity	No activity
Colistin	Variable / resistance common	Variable / resistance common	Variable / resistance common	Variable / resistance common
Tigecycline	Variable / resistance common	Variable / resistance common	No activity	Variable / resistance common

---

B. Gram-Positives

Antibiotic	Staphylococcus aureus	Enterococcus faecalis	Enterococcus faecium
Oxacillin	Reliable activity (MSSA)	No activity	No activity
Vancomycin	Reliable activity	Reliable activity	Variable / resistance common (VRE)
Linezolid	Reliable activity	Reliable activity	Reliable activity
Daptomycin	Reliable activity	Reliable activity	Reliable activity
Clindamycin	Variable / resistance common	No activity	No activity
TMP-SMX	Variable / resistance common	No activity	No activity

C. Anaerobes

Antibiotic	Bacteroides fragilis	Clostridioides difficile
Metronidazole	Reliable activity	Reliable activity
Piperacillin–tazobactam	Reliable activity	Variable / resistance common
Carbapenems	Reliable activity	Variable / resistance common
Clindamycin	Variable / resistance common	No activity

---

2. Resistance Mechanism Map (AMR Logic Layer)

A. ESBL (Extended-Spectrum β-Lactamases)

Affects:

- Escherichia coli
- Klebsiella pneumoniae

Breaks:

- Ceftriaxone
- Cefixime
- Cefpodoxime

Survives:

- Carbapenems
- Ceftazidime–avibactam

B. AmpC β-lactamases

Seen in:

- Enterobacter cloacae

Breaks:

- Piperacillin–tazobactam (variable)
- Cephalosporins

Survives:

- Cefepime

- Carbapenems

C. Carbapenemases

Types:

- KPC
- NDM
- OXA-48

Effect:

- Destroy carbapenems

Solutions:

Enzyme	Drug
KPC	Ceftazidime–avibactam
NDM	Aztreonam–avibactam
OXA-48	Ceftazidime–avibactam

D. MRSA Mechanism

- Altered PBP (PBP2a)

Resistant to:

- Oxacillin
- Penicillins

Active drugs:

- Vancomycin
- Linezolid
- Daptomycin

E. VRE Mechanism

- D-Ala → D-Ala-D-Lac modification

Resistant to:

- Vancomycin

Active:

- Linezolid
- Daptomycin

---

3. Clinical Decision Framework

A. If organism = ESBL-producing E. coli

→ Use:

- Meropenem

→ Avoid:

- Ceftriaxone

B. If organism = MDR Pseudomonas

→ Use:

- Ceftolozane–tazobactam
- Cefepime (if susceptible)

C. If organism = Acinetobacter baumannii (XDR)

→ Use:

- Colistin
- Tigecycline
- Cefiderocol

D. If organism = MRSA

→ Use:

- Vancomycin
- Linezolid

E. If organism = VRE

→ Use:

- Linezolid
- Daptomycin

F. If anaerobic infection

→ Use:

- Metronidazole
- Piperacillin–tazobactam

---

4. Time-Trend Resistance Evolution (Critical Insight Layer)

Phase 1: Penicillin Era

- High susceptibility
- Minimal resistance

Phase 2: β-lactamase Emergence

- Penicillin failure
- Rise of aminopenicillins

Phase 3: Cephalosporin Expansion

- Ceftriaxone dominance
- ESBL emergence

Phase 4: Carbapenem Era

- Gold standard for MDR
- CRE emergence

Phase 5: Modern AMR Crisis

- MDR, XDR organisms
- Reliance on:
    - Colistin

- Tigecycline
- Novel inhibitors

---

1. Enterobacterales (Core Gram-negative Group)

Includes:

- Escherichia coli
- Klebsiella pneumoniae
- Enterobacter cloacae
- Citrobacter freundii
- Serratia marcescens
- Proteus mirabilis
- Morganella morganii
- Providencia stuartii

A. First-Line (Susceptible strains)

- Ceftriaxone
- Cefepime
- Piperacillin tazobactam
- Ciprofloxacin

B. ESBL Scenario

→ Use:

- Meropenem
- Imipenem

C. Carbapenem-Resistant (CRE)

→ Use:

- Ceftazidime avibactam
- Meropenem vaborbactam
- Aztreonam avibactam

D. Species-Level Exceptions

- Klebsiella pneumoniae → intrinsic ampicillin resistance
- Enterobacter cloacae → AmpC inducible
- Serratia marcescens → cephalosporin resistance risk

---

2. Non-Fermenting Gram-Negatives

A. Pseudomonas Group

- Pseudomonas aeruginosa
- Other Pseudomonas spp

Active Drugs

- Cefepime
- Ceftazidime
- Piperacillin tazobactam

- Meropenem
- Ceftolozane tazobactam

MDR Scenario

→ Use:

- Ceftolozane–tazobactam
- Colistin

B. Acinetobacter Group

- Acinetobacter baumannii
- Other Acinetobacter spp

Active Drugs

- Ampicillin sulbactam
- Tigecycline
- Colistin

XDR Scenario

→ Use:

- Colistin
- Cefiderocol

C. Special Case

- Stenotrophomonas maltophilia

→ First-line:

- Trimethoprim sulfa

---

3. Gram-Positive Cocci

A. Staphylococci

- Staphylococcus aureus
- Coagulase-negative staphylococci

MSSA

→ Use:

- Oxacillin

MRSA

→ Use:

- Vancomycin
- Linezolid
- Daptomycin

B. Enterococci

- Enterococcus faecalis
- Enterococcus faecium

E. faecalis

→ Use:

- Ampicillin
- Vancomycin

VRE (E. faecium)

→ Use:

- Linezolid
- Daptomycin

C. Streptococci

- Streptococcus pneumoniae
- Streptococcus pyogenes
- Streptococcus agalactiae

First-line

→ Use:

- Penicillin
- Ceftriaxone

---

4. Anaerobes

Includes:

- Bacteroides fragilis
- Clostridioides difficile
- Prevotella spp
- Peptostreptococcus spp

First-Line Drugs

- Metronidazole
- Piperacillin tazobactam
- Carbapenems

Key Exception

- Clostridioides difficile

→ Use:

- Metronidazole (mild)
- Vancomycin (severe)

---

5. Respiratory / Fastidious Organisms

Includes:

- Haemophilus influenzae
- Moraxella catarrhalis
- Neisseria gonorrhoeae

First-Line

- Ceftriaxone
- Azithromycin

---

6. Key Global Insight (Very Important)

The pattern is:

- **Enterobacterales → plasmid resistance (ESBL, CRE)**
- **Pseudomonas → intrinsic + adaptive resistance**
- **Acinetobacter → extreme MDR accumulation**
- **Gram-positives → target modification (MRSA, VRE)**
- **Anaerobes → generally stable but slowly shifting resistance**